Teams Workflow Setup Guide

Handling AWS Billing Alerts via a Single Webhook

1. How It Works

Instead of firing off multiple requests, our Python script sends the AWS Billing report just once to a single Webhook. From there, Microsoft Teams Workflow catches the incoming Adaptive Card and broadcasts the exact same message to everyone who needs it.

```
Python Script
  ↓ (Single Webhook)
Teams Workflow
  ├── Send to User 1
  ├── Send to User 2
  └── Send to User 3
```

2. Why do it this way?

- It grabs the HTTP request from Python using the trigger: `When a Teams webhook request is received`.
- It extracts the full Adaptive Card using `triggerBody()`.
- It forwards the card to as many people as you want in Teams.
- **The main benefit:** You don't have to create a bunch of different Webhooks for each team member. Plus, you can easily add or drop people right from the Workflow interface without ever touching the backend Python code.

3. Setting Up the Workflow

1. Open Teams and go to the **Workflows** app.
2. Create a new flow and pick the trigger: `When a Teams webhook request is received`.
3. In the trigger settings, make sure to set **Who can trigger the flow** (or Authentication) to **Anyone**.
4. Save it and copy the Webhook URL it gives you.

5. Drop that URL into your `.env` file or Kubernetes Secret as `TEAMS_WEBHOOK_URL`.

4. Sending the Card

Next, you'll need to add an action called: `Post card in a chat or channel`. Here is how to fill it out:

Field	What to put
Post as	Flow bot
Post in	Chat with Flow bot
Recipient	The email or name of the user in Teams
Adaptive Card	`triggerBody()`

> **Watch out:** Don't just type `triggerBody()` normally. You have to select it from the **Expression** tab. Also, make sure it doesn't accidentally get wrapped in an *Apply to each* loop. Since we're sending one card and not a list, a loop will break the flow.

5. Adding More People

Need to add someone else to the loop? Don't touch the Python script. Just duplicate the sending step in your Workflow.

1. Open your Workflow.
2. Find your existing `Post card in a chat or channel` step and click the three dots (**...**).
3. Hit **Copy** (or just add a new action with the same name).
4. Set it up exactly like the others (Flow bot -> Chat with Flow bot).
5. Update the **Recipient** to the new user's name.
6. Set the **Adaptive Card** to `triggerBody()` (again, using the Expression tab).
7. Save it and run `python handler.py` to test.

Here is what it looks like with a fourth user added:

```
Python Script
  ↓ (Single Webhook)
Teams Workflow
  ├── Send to User 1
  ├── Send to User 2
  ├── Send to User 3
  └── Send to User 4
```

6. Removing Someone

1. Open the Workflow.
2. Find the card-sending step for the person you want to remove.
3. Click the three dots (**...**) on their step.
4. Hit **Delete** and then Save. That's it!

7. Testing It Out

Run your script locally to see if it works:

```
python handler.py
```

If everything is good, your terminal should say something like:

```
Teams webhook response: 202
Teams notification sent for account: AWS-DEV
```

You can also check the **Run history** in the Workflows app to verify that each person's step says **Succeeded**.

8. Quick Troubleshooting

What went wrong	How to fix it
Getting a `401 DirectApiAuthorizationRequired` error	Make sure the Trigger Authentication is set to **Anyone**.

What went wrong	How to fix it
Script says 202 Success, but no one got the message	Double-check that you used `Post card in a chat or channel` and not the standard "Post a message" action.
The card shows up blank in Teams	You typed `triggerBody()` as plain text instead of using the **Expression** tab.
Power Automate forced my step into an "Apply to each" loop	Drag your step out of the loop and delete the loop entirely. The card should sit directly under the Trigger.
Only some people got the alert	Go into the **Run history**, open the specific run, and look at the step that failed to see what went wrong with that specific user.

9. The Ideal Setup

```
When a Teams webhook request is received
  ├── Post card in a chat or channel → User 1
  ├── Post card in a chat or channel → User 2
  └── Post card in a chat or channel → User 3
```

The takeaway: Keep the code simple by making it fire once, and let Teams handle the heavy lifting of routing the message to your team.